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From Averages to Atavism: How the Victorian Rejection of Variation Shaped Early Criminal Profiling

A common misconception regarding science is that it is a process of progressing towards truth independent of subjective bias or societal pressures. The Victorian period, a period of fixation on order, categorization, and the notion that human nature could be quantified by numbers, illustrates how science is not always a reflection of truths but a reflection of what society wishes to be true. In the nineteenth century, the doctrine of "biological determinism" emerged. The doctrine argued that one's value, intellect, and even propensity for committing crimes were determined at birth based on his or her genes. "Biological determinism" relied upon a particular rhetorical framework; specifically, it privileged "types" and "averages" above variance. When one compares the arguments made by Stephen Jay Gould with present-day historiography, the Victorian dismissal of variance becomes apparent as not only scientifically inaccurate but a rhetorically-driven attempt to turn prejudiced ideas into indisputable orthodoxy.

An accomplished scientist and prose stylist, Stephen Jay Gould devoted a considerable part of his life to being the historian of these perilous notions. In the *Modern American Prose* book, Gould styles himself as a "Galilean" writer- one who takes delight in "nature's intellectual puzzles and our quest for explanation and understanding" (Gould 207). Gould describes the difference between the "Galilean" and "Franciscan" writing style in his "Prologue" to *Bully for*

Brontosaurus. While the latter refers to nature's aesthetic appeal, the former is used by the Victorians as a means to achieve an aura of authority. By providing some sort of measure, some kind of number, the Victorians thought that they had discovered some laws of nature. Using his shrewd and vigorous style of writing, Gould deconstructs this myth of objectivity and shows how "science... is a creative human activity" (Gould 208). According to him, we need to be ever wary of any findings that conveniently support "social and political prejudices" (Gould 208). The narrative of science is crucial because its power and beauty should be available to the public so that there are "active inquiring citizens" (Gould 213).

Biological determinism originated from the Victorian desire to classify individuals in quantifiable categories. The definition of biological determinism according to the *Encyclopedia Britannica* states that it is the "idea that human behavior is determined by an individual's genes or some physiological characteristic of the individual." During the Victorian Era, this implied that scientists endeavored to determine the average measurements of various racial groups to establish what was "normal." In "Biological Determinism as an Ideological Weapon," Richard Lewontin describes this as not only a scientific error but also an "ideological weapon" used to reinforce the "current order of things" by suggesting that social hierarchy was purely a matter of nature. By focusing on "fixed types," Victorian science overlooked the complexity of human beings and insisted that "particular races are biologically determined to occupy inferior social standings." It was a dismissal of the "humanistic tradition," as described by Gould (Gould 213).

Perhaps the greatest persuasive impact that the idea of an "average" exerts can be best exemplified through Gould's dissection of 19th century craniometry in his essay "Women's Brains." In examining the theories of Paul Broca, a researcher who relied upon averages of brain

weight in concluding that women were inherently intellectually inferior to men, Gould makes clear the ways in which 19th century researchers engaged in the use of statistics in order to lend credence to their culturally motivated theories (Gould 205-214). This is evidenced, according to Stanley Finger and Paul Eling in "The Quest for Objectivity and Measurements in Phrenology's 'Bumpy' History," in the fact that Victorian neurologists were looking for a "veneer of mathematics" in order to prove themselves right. Finger and Eling substantiate Gould's assertion that Victorian neurologists fixated upon the idea of standardized measurement of the human cranium since they believed that the physical "types" would reflect personality traits. By relying upon a single numerical value, the average, Broca was able to overlook the vast range and variation in the human population, effectively twisting one skewed statistic into biological law.

The negation of individual differences made a huge difference in the field of criminal justice, giving rise to the practice of profiling. Rather than trying to understand what makes someone commit a crime, Victorians attempted to identify what the "criminal type" was. In their view, the physical dimensions of one's head or jawline could give away whether an individual was a criminal or not. This was what Cesare Lombroso pioneered, and thus became known as the father of modern criminology. As per *Britannica Editors*, Lombroso was convinced of "atavism," the theory that criminals were evolutionary throwbacks possessing characteristics of primitive man.

The study of Lombroso draws upon the Victorians' infatuation with "types" and creates a type portrait of the "natural-born criminal." In his view, some physical characteristics, known as "stigmata," existed among the criminals, such as wide jawlines, prominent cheekbones, or a lopsided face. Lombroso argues that such physical characteristics do not just happen by chance,

but rather, they are due to biological determinism, and thus, these criminals become biological "tags" for their inevitable deviant behavior. By applying these physical measures, Lombroso converted the practice of criminal detection into a biological experiment. If one person had the average bodily characteristics of a criminal, then it was genetically destined that he will be a criminal. As such, using this method allowed the law to ignore all outside influences that might have been behind his crimes, ranging from societal issues to educational background, and focus entirely on his genetics. This is precisely what Gould defines as the practice of "biological labeling" (Gould 209).

Moreover, the practice of profiling is also associated with the ideology of racial essentialism. According to the writings of W. Carson Byrd and Matthew W. Hughey titled "Introduction: Biological Determinism and Racial Essentialism: The Ideological Double Helix of Racial Inequality," the Victorian age's obsession with classification has created the idea of an inequality double helix. With the help of the notion of "race" or "type," as being a natural phenomenon, 19th century jurisprudence and science have been able to embed the practice of racism within the foundations of investigative profiling as well. It is the rhetoric of profiling that helped the Victorian elite maintain control over the masses while claiming the innocence of their actions through the neutral language of "science." As pointed out by Garland Edward Allen in his article "Biological determinism: theory and its application," the concept helped people sustain the existing "social and economic status quo" by suggesting that poor individuals belong to such a station in life biologically speaking.

But Gould puts forward the strongest rebuttal to this rationale in his essay "The Median Isn't the Message." When confronted with a life threatening condition, Gould analyzed the

"average" figures of life expectancy and realized that this is all just a "simplification", ignoring the person in question. "Variation itself is nature's only irreducible essence" (Gould 214). While the figure of "average" or "median" can give some indication about the situation in general, it says absolutely nothing about the person who is in front of you right now. During the Victorian times, the motto of the society was "average is the message" and if you were different then you were considered as an "outlier" and thus a "mistake". What Gould proves through this essay is that it is the variation which is the real essence of life, while the "average" or "median" is the tool through which scientists try to impose their opinions as objective truths.

This technique that Gould employs in *Modern American Prose* is intended to make these motives known to his readers. For instance, in his "Prologue" in *Bully for Brontosaurus*, Gould justifies the usage of "lateral connections"- connecting the sciences to their history, politics, and culture- to prove that any factual statement about nature lacks isolation. Gould aims at inspiring his readers to be "active inquiring citizens" and not blindly accept the information presented to them (Gould 213). Through uncovering "the flaws in the Victorian veil" (Gould 215), Gould points out how by letting the experts define the "average" and the "type", we are often handing over the power to judge us, thus making us feel valued or worthless. In the essay titled "Flaws in a Victorian Veil," Gould examines the life story of the brilliant scientist, Louis Agassiz. Gould provides strong evidence of the impact that prejudice had on Agassiz, who was one of the most prominent scientists during his time, in denying the fact that all humans are equal (Gould 215-221).

However, in order to comprehend this problem better, it is important to see how the legacy of the Victorian age lives on through us. Even though it is no longer considered

acceptable to profile a person based on his or her kind or genes, it still persists in contemporary times, having shifted its focus from cranium measurements to mathematical calculations.

Although profiling people based on their type or genetics is not an accepted practice anymore, it has not vanished from modern society, moving instead from cranial measurements to statistical analysis and calculations. Predicting crimes based on the data is nothing other than falling into the same "typological" pitfall that once entrapped Lombroso. Instead of analyzing individuals separately, people choose the average. According to Lewontin, it is all too easy for science to turn into a weapon, which needs constant verification of variation. Therefore, the rejection of variation in the Victorian age was a method of escaping the complex truth about the equality of human beings. Making science appear as a list of types allowed the existence of inequality in the world.

Moreover, the "Measured Style" of the Victorian era was a strategy used to create "Ethos." In other words, if a scientist could provide a graph or table of averages, then they would be considered infallible. In contrast, Gould utilizes "Pathos," an appeal to emotion through the use of his personal experience in his essay "The Median Isn't the Message," to point out the humanity in numbers. This means he exhorts readers to "rage mightily against the dying of the light" (Gould 214), a statement that refers to the fight against the deterministic nature of "death sentences" provided by numbers. In effect, Gould makes the connection between the scientist and the layperson by appealing to their humanity, something that the Victorians deliberately avoided.

As the essay "Evolution as Fact and Theory" notes, the scientific concept of truth is never complete. Instead, it is an ongoing process of "checking and correcting" (Gould 231). The

Victorians, on the other hand, desired their "types" to be eternal and immutable. They sought "the standard type of the criminal," "the standard type of woman," and "the standard type of laborer." The need for eternity is one characteristic of biological determinism, but it is contrary to nature's principles. Evolution thrives on variability; without variability, there would be no evolution, no adaptation, no life. Thus, the Victorian denial of variability is not only a social error; it is biologically incorrect.

In summation, the rhetoric surrounding "types" and "averages" in Victorian culture was ultimately a failure that produced decades of biological determinism and flawed criminal profiling based on averages. In overlooking the "ultimate reality" of variation, the scientists of the nineteenth century created an era in which people were judged according to the contours of their heads and not the quality of their conduct. As Gould explains, the "true message" of nature is not to be found in its singular median point or "standard type," but rather in the wide spectrum of human variation and diversity that is both unpredictably beautiful and endlessly fascinating. In order to maintain a free society, however, the "myth of objectivity" must continually be scrutinized, and we should always resist the "system" insofar as it attempts to determine us according to our "average." After all, science is itself a human endeavor and will remain as such as long as it is practiced by humans.

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